Project Pitch

Human Need: As the focus of our design project is reducing household food waste, the human need we identified is "I want to reduce my household food waste". This is a problem that many users face when they have an excess of groceries and food products approaching the expiry date. Stakeholders include people who cook for themselves, the food insecure, waste management services, and the environmentally conscious, as they all handle household food waste.

Reference Designs: We focused our reference designs on apps targeted towards reducing food waste, specifically Supercook and Too Good to Go. Supercook allows users to select ingredients and generate recipes that utilize those ingredients to prevent food waste, while Too Good To Go allows restaurants and businesses to sell food that is about to be thrown out for a reduced price.

Initial Considerations: By critiquing the reference designs, we developed initial considerations our design should address. After evaluating Supercook, we decided that our considerations should include "flexible options", "error correction", and "visual clarity". By analyzing Too Good to Go, we settled on "efficiency", "visual cues", and "user interaction".

Evaluation Research Questions

- RQ1: How often do you buy an excess of groceries that is ultimately wasted?
- RQ2: What type of food is wasted the most in your household and why?
- RQ3: What approach(es) do you take to reduce your food waste?

Fieldwork Study Protocol

For the full fieldwork study protocol, see Appendix A.

Participant Inclusion/Exclusion Criteria: The only criteria is that all participants will either be a) 19+ or b) 17+ if they are a student currently attending the University of British Columbia.

Participant Sample Description: We recruited participants from a variety of living situations, including participants who lived on their own, with roommates, or with their families. All participants were recruited from CPSC 344 Section 102, with ages ranging between 17-30.

Protocol: This research study was conducted as an in-person study using a demographics questionnaire followed by an interview. The interviewer first explained the purpose of the study and how the study would run before proceeding to the demographics survey and interview. The demographics survey also contained the consent form for the study. The interview was conducted as a semi-structured interview using questions outlined in the interview script. The interviewer recorded responses on each participants' datasheet and asked follow-up questions if necessary. *Please note that our protocol was initially designed as a mixed-methods research study using an interview and questionnaire. After discussion with the teaching team, we chose to use just an interview because the questions significantly overlapped between the two methods.*

List of Collected Data: Using this research approach, we primarily collected qualitative data. Questions 1 and 2 in the questionnaire provided categorical information regarding participants' demographics (age range and their living situations). Questions 1-8 in the interview collected free form responses for thematic analysis pertaining to food waste habits outlined in RQ1, RQ2, and RQ3.

Results

RQ1: How often do people buy excess groceries that are ultimately wasted?

- CLAIM: Most people buy an excess of groceries that are ultimately wasted once a month or less.
- EVIDENCE: We interviewed 11 people who either purchase groceries or assist with purchasing groceries for their household about how often they throw out food or groceries. Out of this sample population, 10 people reported that they buy an excess of groceries that become wasted. 7 people reported they threw out their groceries once a month or less, 3 people reported that they threw out their groceries once every one to two weeks, and 1 said they never wasted food. This is also represented in Figure 1: Number of people who said their food gets wasted "Never", "<= 1 time / month" or "1-2 times / month" from the interviews.
- CONFIDENCE: Low. Our confidence level is limited by a reliance on self-reporting of food waste, which may cause a bias and won't accurately reflect participants' true experiences, potentially skewing the data. Additionally, our small sample size may not be representative of the entire population, as we only interviewed university students.

RQ2: What type of food is wasted the most in your household and why?

- CLAIM: Most households tend to waste items with short shelf life (of a few days). Of these items with short shelf life, vegetables, fruits and prepared meals are the most commonly wasted mainly because of quick food expiration, buying more than is needed, or forgetting about the food completely.
- EVIDENCE: One of our interview questions was "Is there a particular type of food item that usually ends up wasted in your household?". Affinity Diagrams 1 & 2 revealed that 8/11 participants reported that items with a short shelf-life were being wasted the most, including instances of vegetables, fruits, prepared meals, meat as well as quickly perishable items. Additionally, the reasons for food waste found in the diagrams mainly included forgetting about the items, estimating the right quantity of items to buy, and the speed of food expiration, with other popular reasons including unwillingness to eat particular items and inadequate storage.
- CONFIDENCE: Medium. Our confidence level is limited by our small sample size, which might not be representative of the entire population as we only interviewed university students. However, we did get confident and stable answers through the interviews.

RQ3: What approach(es) do people take to reduce their food waste?

- CLAIM: The solutions that participants employ to reduce their food waste can be categorized into four themes on groceries/food: estimating food buying quantity, proper food storage/preservation, food usage or consumption, and tracking.
- EVIDENCE: One of our interview questions was "What strategies do you try to reduce your food waste?". Affinity Diagram revealed the four categories of strategies/solutions above utilized by the 11 participants.
- CONFIDENCE: Medium. Due to the small sample size (n = 11), we were unable to get more variable answers as most of the participants were from our CPSC 344 class. We did get a variety of answers for this one, but we feel like we could get more variations for a bigger sample size.

Plots, Diagrams and/or Tables

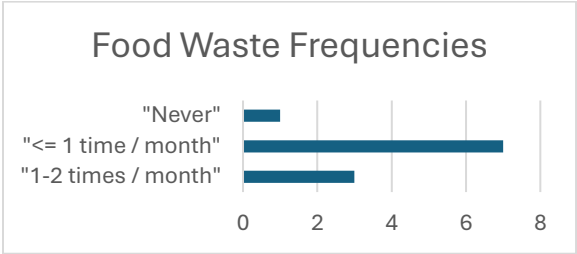
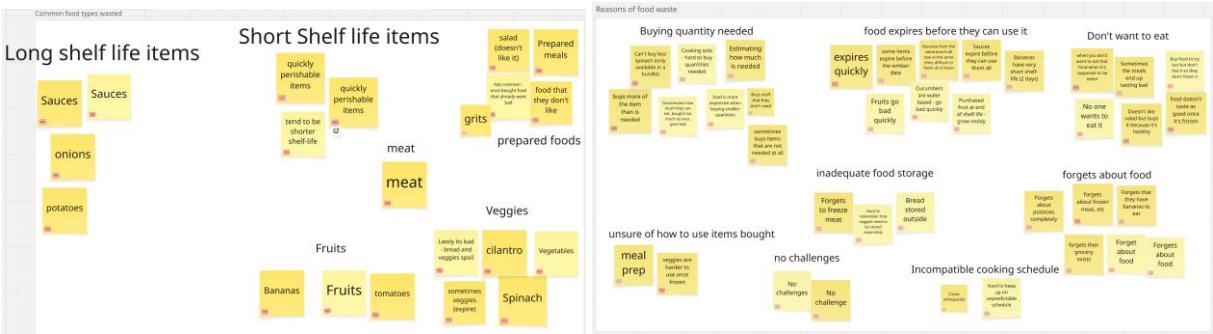
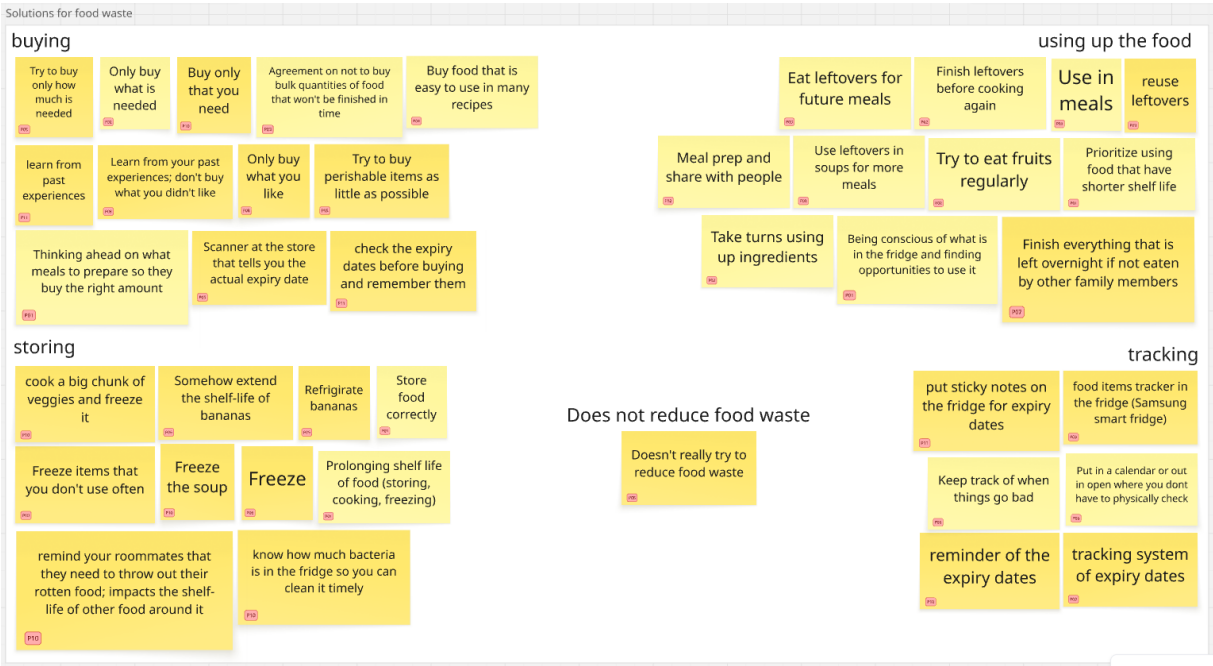


Figure 1: Number of people who said their food gets wasted "Never", "<= 1 time / month" or "1-2 times / month" from the interviews



Affinity Diagrams 1 & 2: Instances of different types of food wasted in participants' households and reasons from the interviews grouped and themed



Affinity Diagram 3: Instances of solutions to food waste given by participants based on the challenges they face from the interviews grouped and themed

User Scenarios

User Scenario 1: Jimothy is a struggling single dad of two who can barely afford to feed his children on his salary. Jimothy buys groceries in bulk because they are cheaper than buying smaller quantities. He wants to reduce the amount of food wasted so he can make his groceries last longer and save money. **Because he wants to use all the groceries he buys to save money, Jimothy needs to track when the groceries expire so that he can use them up beforehand.**

Task Analysis - User 1

- go to the grocery store
- browse around the aisles with a shopping cart
- check the price of food products and their expiry date, avoiding expired products
- put desired grocery items into shopping cart
- check out
- return home
- place grocery items into fridge
- write down list of expiry dates of purchased items
- refer to the list of expiry dates to decide which ingredients to use in future cooking
- cross fully used ingredients off the list

User Scenario 2: Ted and Andre are brothers living together and university students. Ted and Andre want to cook less-mainstream cuisines for themselves to give their meals more variety. However, when they buy uncommon ingredients, the unused portion just gets wasted. **Ted and Andre want to find a way of using leftover ingredients to reduce their food waste.**

Task analysis - User 2

- have an interesting dish in mind
- find a recipe for its ingredient list
- go to the grocery store
- find ingredients
- put ingredients into shopping cart
- checkout groceries
- take groceries home
- prepare two servings of food according to the recipe
- review remaining ingredients left unused
- search for uses for remaining unused ingredients or store them for future use

Central Tasks

Central Task [1]: Track Food Inventory

- A user needs to be able to keep track of what foods they have, how much of that food they have, and when it will expire
- Justification: Our interviews discovered that many participants reported the reason for food waste as being a failure to track food inventory. 5/11 participants reported difficulty using their ingredients before the expiry date, while 6/11 participants reported forgetting about their groceries entirely.

Central Task [2]: Find Use Ideas

- A user needs to find a way of using the ingredients before it expires
- Justification: During our data collection process, we discovered that 5/11 participants suggested to be more conscious about using up their groceries using various meal preparation methods, as a solution to reducing household food waste. Many people reported not knowing what to do with the groceries, which resulted in their groceries becoming unused and eventually spoiling.

Requirements

Requirements for Central Task [1] Track Food Inventory

- **Design Constraint A – Itemization** (functional constraint)**:** The design must present a list of ingredients and their associated expiry dates with which to track the user's groceries, that can be added to and subtracted from at the user's discretion.
  - **Justification:** 5/11 participants in our study mentioned expiry date as a reason for wasting food, 6/11 mentioned completely forgetting about food items, and 5/11 mentioned tracking expiry dates as a solution to their food waste.
  - **Metric A1** (evaluation-based): 90% of users (out of a sample of 10 test participants) must report either "Agree" or "Strongly Agree" on a 5-point Likert scale when asked about their satisfaction with the design's grocery list functionality.
- **Design Goal B** – **Reminding:** The design should improve the user's awareness of information as it becomes urgent, prioritized by order of urgency.
  - Justification: From our study's data, we discovered that 6/11 participants reported the reason for their food waste was forgetting about their food/groceries.
  - **Metric B1** (evaluation-based): # of participants who report using grocery/food items that would have otherwise been forgotten due to our design (more is better)

Requirements for Central Task [2]: Find Use Ideas

- **Design Constraint C – Dietary Restrictions** (user characteristic constraint): The design should support users with dietary restrictions when generating recipes and usage ideas.
  - **Justification:** Our consideration of "flexible options" drawn from our critique of Supercook would be impacted negatively if the option to filter for dietary restrictions did not exist. Additionally, one participant mentioned being vegetarian and thus having different experiences with food items wasted compared to other participants who did not indicate dietary restrictions.
  - **Metric C1** (evaluation-based): Recruit 10 users of varying dietary restrictions to try the design and report if the search results were tailored to their dietary restrictions. At least 9 users must have a satisfactory response.

General Requirements (Applicable to both tasks)

- **Design Goal D – Visibility:** Users should be able to discern and access relevant information and features quickly and easily.
  - **Justification:** From our reference design critiques, we found visual clutter to be a negative factor toward our considerations (see "Initial Considerations").
  - **Metric D1** (evaluation-based): # of elements a user can identify in a short period of time when looking at certain screens/pages (more is better)
- **Design Goal E – Efficiency:** Users should be able to interact with frequently used features with minimal intermediate steps.
  - **Justification:** Our consideration of "efficiency" greatly impacts the other considerations of "flexible options" and "user interaction", placing it at high priority.
  - **Metric E1** (heuristic-based): # of interactions required to input a new grocery item to the list (fewer is better)

- **Metric E2** (evaluation-based): amount of time required to find a concrete use for urgently expiring ingredients (shorter is better)

Appendix A
Evaluation Instruments

The following documents can be found in this shared UBC OneDrive folder: Appendix A

- Full Fieldworks Study Protocol
  - Interview Script
  - Questionnaire
- Consent Form (Qualtrics Questionnaire is in Questionnaire PDF)
- Call for participation form
- Affinity Diagram: https://miro.com/app/board/uXjVLi_qKZc=/?share_link_id=533202334187

Appendix B
Additional Documentation

The following documents can be found in this shared UBC OneDrive folder: Appendix B

- Signed "Adherence to Ethics Protocol" form
- Raw data from:
  - Interview Script
  - Questionnaire and Survey Results
- Signed participant consent forms
  - *Please note that the signed participant consent forms are located as responses in the questionnaire*